

# prepare\_spatial\_variance\_waffle\_data.R

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prepare\_spatial\_variance\_waffle\_data  
*Prepare Spatial Variance Waffle Data*

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## Description

Creates tile positions and precomputed trivariate fill colours for unit-level waffle plots.

## Usage

```
prepare_spatial_variance_waffle_data(  
  data_plot,  
  vec_component_colours,  
  n_waffle_rows = 5L,  
  vec_required_components = base::c("Abiotic", "Spatial", "Associations"),  
  ranking_column = "component_total_percentage",  
  anchor_component = "Associations",  
  method = base::c("HCL", "perc_avg")  
)
```

## Arguments

|                                      |                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <code>data_plot</code>               | Data frame returned by [prepare_spatial_variance_plot_data()] and containing variance components for each observation.   |
| <code>vec_component_colours</code>   | Named character vector mapping canonical component IDs to base HEX colours.                                              |
| <code>n_waffle_rows</code>           | Positive integer giving the number of rows in each waffle panel.                                                         |
| <code>vec_required_components</code> | Character vector of required canonical component IDs used for trivariate colour mixing.                                  |
| <code>ranking_column</code>          | Single character string naming the column used for within-panel tile ordering. Defaults to 'component_total_percentage'. |

`anchor_component`

Optional single component ID used to select one row per observation after colour mixing. Defaults to "Associations" to preserve historical behaviour. When 'NULL', the first value in 'vec\_required\_components' is used.

`method`

Character string selecting the colour-mixing method passed to `[compute_variance_component_colour]`

## Value

A tibble with one row per observation and columns 'tile\_col', 'tile\_row', 'tile\_fill\_colour', and 'point\_colour'.

## Examples

```
data_plot <-
  tibble::tibble(
    scale = base::rep("local", 6),
    scale_id = base::c(
      "local_01",
      "local_01",
      "local_01",
      "local_02",
      "local_02",
      "local_02"
    ),
    resolution_id = base::rep("genus", 6),
    resolution_label = base::rep("Genus", 6),
    component = base::c(
      "Abiotic",
      "Spatial",
      "Associations",
      "Abiotic",
      "Spatial",
      "Associations"
    ),
    continent_id = base::c(
      "europe",
      "europe",
      "europe",
      "asia",
      "asia",
      "asia"
    ),
    component_total_percentage = base::c(50, 30, 20, 30, 30, 40)
  )

prepare_spatial_variance_waffle_data(
  data_plot = data_plot,
  vec_component_colours = base::c(
    "Abiotic" = "#D95F02",
    "Spatial" = "#7570B3",
    "Associations" = "#1B9E77"
  )
)
```

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